LECTURE ACTIVITY 4: LET'S MAKE A DEAL! – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is based on one created by the STAT 1110 Course Coordination Team, with thanks

Learning Objectives:

By the end of this activity, you should be able to

1) Conduct simulations to approximate probabilities.
2) Identify the need to increase the number of repetitions in a simulation on the accuracy of an approximate probability.
3) Identify and interpret probability as a long-run proportion.
4) Select an appropriate decision involving a random process, based on probabilities and/or simulation results.
5) Describe and identify the parameter of interest based on the description of a random process or population.

Key Terms:

- Random process
- Simulation
- Probability

Part I: Random Processes

Let's Make a Deal is an ongoing TV game show that originated in the 1960s. The original premise of the show was relatively simple: contestants would make a series of deals with the host - the most famous host being Monty Hall - with the hopes of receiving exciting prizes. Monty would present the contestants with three doors: behind one of them is a "good" prize like a car, a vacation, electronics, etc., but behind two of them is what was known as a **Zonk** (essentially a "bad" prize), like a live goat or a worthless novelty item[1].

Contestants would then select one door they believe to contain the good prize. But, there was a twist - instead of revealing the contents of the door the contestant picked, Monty would instead reveal a Zonk behind one of the two doors the contestant didn't pick. The contestant would then be presented with two options: **stay** (i.e. keep the contents behind the door they originally picked) or **switch** (i.e. keep the contents behind the unopened door).

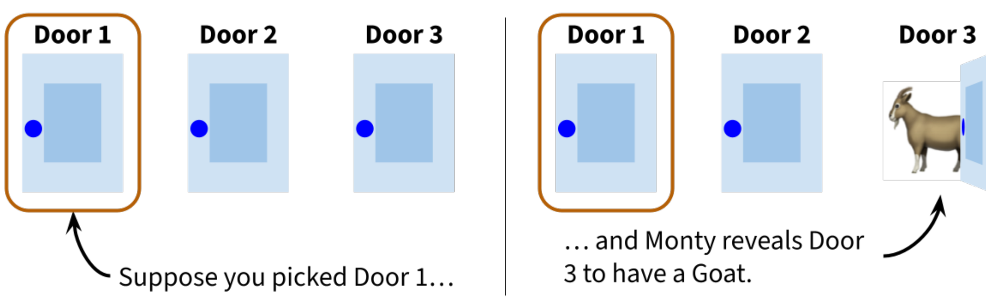

Figure 1: Example game play in which you pick Door 1 and Monty reveals the contents of Door 3.

[1] https://lets-make-a-deal.fandom.com/wiki/Zonk

Question 1

Based on your intuition alone, do you think there is a strategy that is better?

- ◯ STAY strategy is better
- ◯ SWITCH strategy is better
- ◯ The two strategies are equivalent

Solution: Answers may vary!

Assuming there is no set pattern to where the good prizes and Zonks are placed each time the game is played, this game is an example of a **random process**: we don't know exactly how each iteration (one play of the game) will pan out, but we do expect to see some patterns in the long run.

With the power of technology, we can actually examine the "long run" empirically, by way of yet another applet. The applet we'll be using is hosted here (for those of you with printed copies of this handout: `http://www.rossmanchance.com/applets/2021/montyhall/Monty.html`).

We'll start by getting a feel for the STAY strategy.

Question 2

(a) Navigate to the applet linked above, and play the game 10 times using the **STAY** strategy. Follow the steps below:

1) Click on a door

2) Click on the same door again (this is because we're starting with the STAY strategy). The computer will reveal whether you won or lost.

3) Write down the outcome (car or goat) in the table below).

4) Press the PLAY AGAIN button and repeat until you have played 10 times.

Answers will likely vary.

Game #	1	2	3	4	5	6	7	8	9	10
Outcome (car or goat)	Goat	Goat	Car	Goat	Car	Goat	Goat	Car	Goat	Car

⚠ Caution

For now, ignore the boxes that say "Repeat X strategy Y times" and follow the instructions above.

(b) Check with your fellow group-mates; did everyone get the same string of outcomes? Did any two people get the same *proportion* of cars and/or goats?

Solution: Answers may vary. I suspect it's very likely that you all had different sequences of outcomes, but some of you may have had similar (or even the same) overall proportion of goats and cars.

Now we'll take a look at the SWITCH strategy.

Question 3

(a) Navigate to the applet linked above, and play the game 10 times using the **SWITCH** strategy. Follow the steps below:

1) Click RESET (so that you start from scratch)

2) Click on a door. The computer will open a door and show you a goat.

3) Click on the **other door** which has the word "switch" written on it; the computer will then reveal whether you won or lost.

4) Write down the outcome (car or goat) in the table below).

5) Press the PLAY AGAIN button and repeat until you have played 10 times.

Answers will likely vary.

Game #	1	2	3	4	5	6	7	8	9	10
Outcome (car or goat)	Car	Car	Car	Car	Car	Goat	Goat	Goat	Car	Car

⚠ Caution

For now, ignore the boxes that say "Repeat X strategy Y times" and follow the instructions above.

(b) Check with your fellow group-mates; did everyone get the same string of outcomes? Did any two people get the same *proportion* of cars and/or goats?

Solution: Again, answers may vary. I suspect it's very likely that you all had different sequences of outcomes, but some of you may have had similar (or even the same) overall proportion of goats and cars.

Question 4

Based on the preliminary findings of Questions (2) and (3) above, does it look like there is any advantage to one stragey over the other? Justify your answer.

Solution: For the vast majority of you, I'd wager you said the SWITCH strategy seems better because it yielded a higher proportion of wins than the STAY strategy.

Of course, in the real game, we only get to play once - we don't get to compare our strategies over multiple plays! But this is one of the beauties of **simulations**; we're able to leverage theory and technology to things we wouldn't be able to do in "real life." In fact, we'll work our way up to testing out the the STAY and SWITCH strategies under *thousands* of plays of the game!

First, let's familiarize ourselves with a new mechanism in the applet.

Question 5

Navigate back to the applet and do the following:

1) Click RESET (so that you start from scratch)
2) Ensure that the window in the middle reads "Simulate strategy **Stay 10** times."
3) Make sure that the ANIMATE option is **unchecked**.
4) Click GO; this will run the simulation 10 times.
5) **DON'T CHANGE ANYTHING ON THE APPLET!** Write down your percentage of wins in the second column of the table below.
6) Again, without changing anything, press GO again. You should now have a total of 20 games completed. Record the percentage of wins among these 20 games in the third column of the table below.
7) Continue pressing GO and recording the percentage of wins in the appropriate columns in the table below until you complete 100 games played.

Answers will likely vary.

Number of Games	**10**	**20**	**30**	**40**	**50**	**60**	**70**	**80**	**90**	**100**
Percentage of wins	**50%**	**35%**	**40%**	**33%**	**30%**	**32%**	**31%**	**31%**	**31%**	**30%**

Question 6

Based on your completed table above, what do you notice about how the percentage of wins changes as you play more games? Does this percentage apepar to be approaching some common value?

Solution: Yup; this percentage does appear to be approaching some common value!

Why do we care so much about whether these proportions approach some common value? Well, because we define a **probability** of an event to be the **long-run proportion** of times that the event would occur if the random process were repeated a very large number of times under identical conditions. Technically, to calculate the probability exactly we would need an infinite number of repetitions - which is not possible to do! As such, we typically *approximate* probabilities by simulated the process many times and obtaining an *empirical* estimate of the probability, by examining the proportion of times that the event occurs in the simulated repetitions of the random process.

Pro Tip

For this course, we'll typically report probabilities as decimals (i.e. as a number between 0 and 1). Other conventions exist, though!

To that end, let's bump up the number of games we play to a thousand, and compare the two strategies (STAY or SWITCH).

Question 7

Navigate back to the applet and do the following:

1) Click RESET (so that you start from scratch)
2) Ensure that the window in the middle reads "Simulate strategy **Stay 1000** times."
3) Make sure that the ANIMATE option is **unchecked**.
4) Click GO; and let the simulation run (this may take a few seconds). Note: if you forgot to uncheck the ANIMATE option, this will take much longer to run.
5) **DON'T CHANGE ANYTHING ON THE APPLET!** Change the "Stay" option to "Switch" (near the middle of the applet, so that it now reads "Simulate strategy **Switch 1000** times") and click GO.
6) The simulation should now continue running, and adding onto the plot it made at the end of the 1,000 simulations of the STAY strategy.

Once you've completed the steps above, fill out the table below to record your proportion of wins and losses under the two strategies. Also, **include a screenshot of the graph from the applet** (or make a sketch if you printed the activity). Your graph should have two sets of results (one blue set for the STAY strategy and one green set for the SWITCH strategy).

Outcome	Stay	Switch
Wins	329 (33%)	654 (65%)
Losses	671 (67%)	346 (35%)
Total	1000	1000

Include your Screenshot or Sketch Here:
Answers will likely vary.

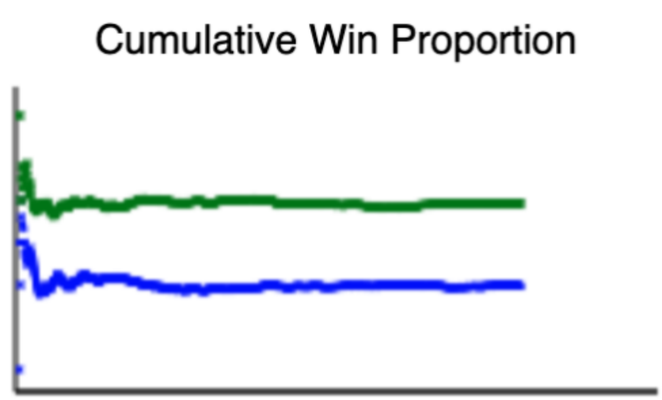

Figure 2: Example graph from the applet

Question 8

Use your results from above to fill in the blanks:

- The approximate probability of winning a game under the STAY strategy is **0.33**
- The approximate probability of winning a game under the SWITCH strategy is **0.65**

Answers will likely vary.

Does one strategy appear to be better than the other? Justify your answer.

Solution: The **SWITCH** strategy seems to be better, because it affords a higher probability of winning.

Spotlight

Figure 3: A photo of Marilyn Vos Savant.
Image Source

Marilyn vos Savant is a brilliant columnist, who, fun fact, had the highest recorded IQ as recorded by the Guinness Book of World Records (before they stopped recording that category) - a whopping 228![a]

So what's her connection to today's activity? Well, the problem of whether to switch or stay is a famous problem from the world of Statistics known as the **Monty Hall Problem** (named after Monty Hall, the original host of *Let's Make a Deal!*). The problem was popularized when it was asked in Marilyn's newspaper column *Ask Marilyn*, in which she showed the correct probability of winning under the SWITCH strategy is 2/3.

Another famous mathematician, Paul Erdős, was skeptical of the answer to the Monty Hall Problem and was only convinced after seeing the answer in simulations[b] - just like we did in today's activity!

[a] https://books.google.com/books?id=qugCAAAAMBAJ&pg=PA36
[b] https://www.mwsug.org/proceedings/2010/stats/MWSUG-2010-87.pdf

Question 9

Explain what it means to say that the probability of winning equals 2/3. **Hint:** recall our earlier definition of probability. I want an *interpretation* of this number, not simply an evaluation statement of whether you think it is large or small.

Solution: If I were to continue playing this game over and over again, using the switch strategy, on the long-run I would expect to win the car in about 66.67% (2/3) of games played.

Part II: Random Processes and Parameters

We spent quite a bit of time in the last Lecture Activity talking about how to deal with data that are samples from a population. Sometimes, however, our data is less about samples from a population and more about observations from a random process. (Recall that a random process is a repeatable process with unknown individual outcomes but a long-run pattern.) Under the latter viewpoint, a parameter can then be re-interpreted as a long-run numerical summary of the underlying process.

Let's get some practice with distinguishing these two "types" of data.

Rock, Paper, Scissors

Welcome to your new life as a professional Rock, Paper, Scissors champion! Now that you've climbed your way to the top, you'd like to make sure you're not showing a tendency to pick rock more than paper or scissors. In a set of 18 games, you chose rock 8 times.

Question 10

(a) How can you describe this study?

- ◯ Sample from a Population
- √ **Observations from a Random Process**

(b) What is the statistic you can calculate from the sample?

- ◯ The sample mean
- √ **The sample proportion**

(c) Calculate the value of the statistic.

Solution: The value of the statistic is $\hat{\pi}$ = **8/18 = 0.4444**, which represents the proportion of times among the 18 games I chose rock.

Marine the Dog

In 2011, an article published by the medical journal *Gut—An International Journal of Gastroenterology and Hepatology* (Sonoda et al.) reported the results of a study conducted in Japan in which a dog was tested to see whether she could detect colorectal cancer by smelling breath and stool samples from patients. There was 1 cancer sample, and 4 non-cancer controls. Marine, an eight-year-old black Labrador, completed 33 attempts of this experimental procedure. Each time, whether or not she correctly identified the bag with the breath of the cancer patient was recorded. In her 33 attempts, Marine correctly identified the bag containing the breath of the cancer patient 30 times.

Question 11

(a) Explain why the data actually represents observations from a random process, and not samples from a population.

Solution: If we thought our data represented samples from a population, we'd need to identify the population. But what's the population here? Note that we are NOT interested in determining the true proportion of people who have colorectal cancer. Rather, we're trying to determine whether Marine *can identify* people who have colorectal cancer.

(b) What is the statistic you can calculate from the sample?

- ◯ The sample mean
- √ **The sample proportion**

(c) Calculate the value of the statistic.

Solution: The value of the statistic is $\hat{\pi}$ = **30/33 = 0.909**, which represents the proportion of times among the 33 attempts Marine correctly identified the bag with cancer breath.

(d) What is the parameter of interest? Remember: because our data is observations from a random process, our notion of a parameter changes from the one we discussed in the last lecture activity! As a hint: the parameter is the long-run proportion of ...

Solution: The parameter is the dog's long-run proportion of correct identifications.

Cal Poly Textbooks

Suppose I wanted to estimate how much money Cal Poly students tend to spend on textbooks in a quarter, but I only have time to ask the 35 students in this statistics class how much they spent.

Question 12

(a) How can you describe this study?

- √ **Sample from a Population**
- ◯ Observations from a Random Process

If you think this is a sample from a population, describe the population and parameter of interest below. If you think this represents observations from a random process, describe the parameter of interest below.

Solution: The population is the set of all Cal Poly students, and the parameter of interest is μ, the average amount of money spent across all Cal Poly students.

(b) Describe the sampling frame. How does it compare to the population - is it bigger, smaller, or the same?

Solution: The sampling frame is the set of all 35 students in this class, and is therefore smaller than the population.

(c) What is the statistic you can calculate from the sample?

- ◯ The sample mean
- √ **The sample proportion**